

ExudedVestibule: Enhancing Mid-air Haptics through Galvanic Vestibular Stimulation

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ABSTRACT

This study presents a novel system that enhances air cannon tactile perception using synchronous galvanic vestibular stimulation (GVS). We conducted a user study with a within-subjects design to evaluate the enhancement effects of synchronous GVS on air cannon tactile sensations across multiple body locations. Results demonstrated significant improvements without affecting the magnitude of physical body sway, suggesting potential applications in virtual reality, particularly for augmenting existing air vortex ring haptics use cases.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

galvanic vestibular stimulation, mid-air haptics, force illusion

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1 INTRODUCTION

Virtual Reality (VR) has emerged as a significant research domain, with numerous investigations delving into various aspects of immersive experiences. Haptic feedback plays a vital role in developing realistic and engaging interactions within VR environments. Conventional haptic feedback systems often involve applying force directly to the user's body, which is limited and uncomfortable. As a result, researchers have been exploring alternative methods for

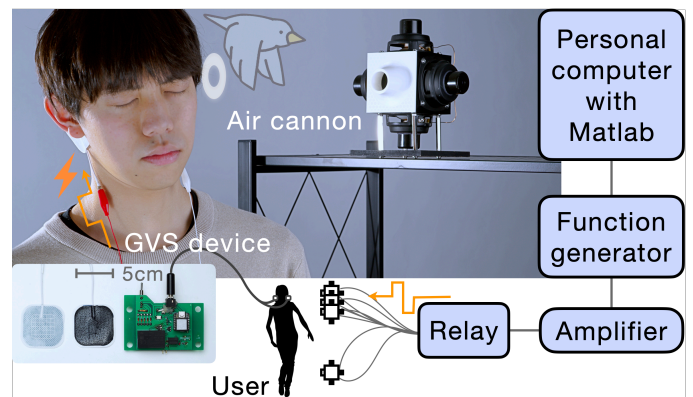


Figure 1: Our technique enhances the user's local air tactile sensation by adding proprioception induced through galvanic vestibular stimulation. It can be applied to multiple parts of the body while wearing a compact device.

providing haptic feedback with minimal force application, leading to advantages such as reduced weight and power consumption.

One promising technology is mid-air haptics, which enables dynamic and localized external force application to multiple body areas without direct contact (for example, [Sodhi et al. 2013]). However, the force sensations produced by existing mid-air haptic systems are relatively weak, limiting their effectiveness in VR environments. Enhanced mid-air haptic systems, such as those by Morisaki et al. [Morisaki et al. 2022], often necessitate device attachments to each target body part. Our study aims to address this limitation by combining mid-air haptics with galvanic vestibular stimulation (GVS) to enhance air tactile sensations. GVS is a well-established technique for inducing proprioceptions by stimulating the user's vestibular system via electrical currents applied between their mastoid processes. Teo et al. [Teo et al. 2020] propose a combination of GVS and external force; however, their work does not specifically focus on mid-air haptics.

2 SYSTEM AND USER STUDY

We developed a prototype system that combines air cannon stimulation with GVS (Figure 1), utilizing air cannons for their ability

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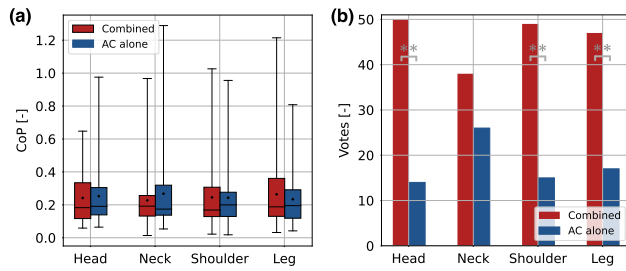


Figure 2: (a) Box-whisker plot of CoP (center of pressure) displacement ($p > 5\%$); white dots: mean, black lines: median. (b) Bar graph showing enhanced tactile sensations with AC + GVS ($p < 1\%$, denoted by **).

to provide strong mid-air haptic feedback over a distance while maintaining simplicity. The system is designed to synchronize the arrival of air cannon stimulation and GVS with a precision of ± 2 ms, ensuring that both stimuli reach the user simultaneously.

Our GVS device's circuit, designed based on [Matsumoto et al. 2021], consists of a constant-current circuit, an H-bridge circuit, and a microcontroller with a 10 bit D/A converter (48 MHz). Square-wave pulses are used for accurate stimulation timing, with a pulse width of 600 ms. A large electrode pad (5 cm x 5 cm, UltraStim X Blue USBX5050, Axelgaard) minimizes pain. The current density at 3.0 mA is 1.2 A m^{-2} . The GVS device weighs 405.0 g, with the battery accounting for 305.0 g. Replacing it with a 5V lithium-ion battery can further reduce the weight. The air cannon device consists of a 3d-printed housing, nozzle, and five 3 inches speakers wired in series and positioned 50 cm away from the user. A bipolar square pulse wave (100 ms pulse width, 200 ms pulse period) drives the speakers (15 V amplitude per speaker). We prepared four air cannons, and mechanical relays switch between each. For further specifications and principles, refer to [Kojima et al. 2023].

The user study employed a within-subjects design, with participants experiencing both combined stimulation (AC + GVS) and control (AC alone) conditions for each body location (right side of head, neck, shoulder, and leg) in a controlled laboratory setting. Measurements included lateral CoP (center of pressure) displacement recorded by a force plate (Flexel) [Yoshida et al. 2022] and paired comparison judgments of perceived tactile intensity for each location. One data set with a value more than twice the size of another was excluded as an outlier. To minimize order effects, the sequence of body locations was counterbalanced. Data analysis involved a sign test for paired comparison judgments and a Wilcoxon signed-rank test for CoP displacement. The user study involved eight healthy adults (mean age 23.0 years, SD = 1.4 years; one female; all right-handed; mean height 169.8 cm, SD = 7.1 cm; mean weight 62.3 kg, SD = 10.3 kg). The experimental protocols were approved by the local ethics research committee at the University of Tsukuba and adhered to the ethical standards of the Declaration of Helsinki.

Participants followed a streamlined experimental procedure that involved briefing, GVS calibration (with resistance $< 30 \text{ k}\Omega$ and maximum current 3.0 mA), maintaining a specific standing posture, receiving multiple paired trials for each body location in random

order, and indicating perceived intensity. CoP displacement data were recorded, and participants answered a questionnaire after the experiment.

Results showed that there were no significant differences ($p < 5\%$) in CoP displacement outcomes between the combined stimulation and control conditions for each body location (Figure 2a), indicating that the physical magnitude of stimulation has not changed. Interestingly, the participants reported significantly stronger tactile sensations in the combined stimulus (AC + GVS) than the single air cannon stimulus (AC alone) in all body locations except for the neck ($p < 1\%$) (Figure 2b). Regarding the questionnaire, participants' interpretations of the cause of the proprioception varied: four attributed it to GVS, one to the air cannon, and three to both.

3 CONCLUSION AND FUTURE WORK

Our user study demonstrates that combining GVS with air cannon haptic feedback enhances localized tactile sensations across multiple body locations without affecting users' physical balance. This approach is a promising first step in increasing the realism and engagement of VR experiences while maintaining user comfort and safety through low-level external stimulation. To determine whether proprioception or electrotactile sensations (side effects) of GVS contribute to the enhancement, future work should implement a sham condition with skin-only electrical stimulation. Additionally, exploring air cannon stimulation through clothing materials, in different postures, at various body locations simultaneously, and integrating our method with other modalities could enrich existing VR applications with air cannon haptics, such as those by Sodhi et al.

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